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Usn = 2VX24UE057

Exp no = 1

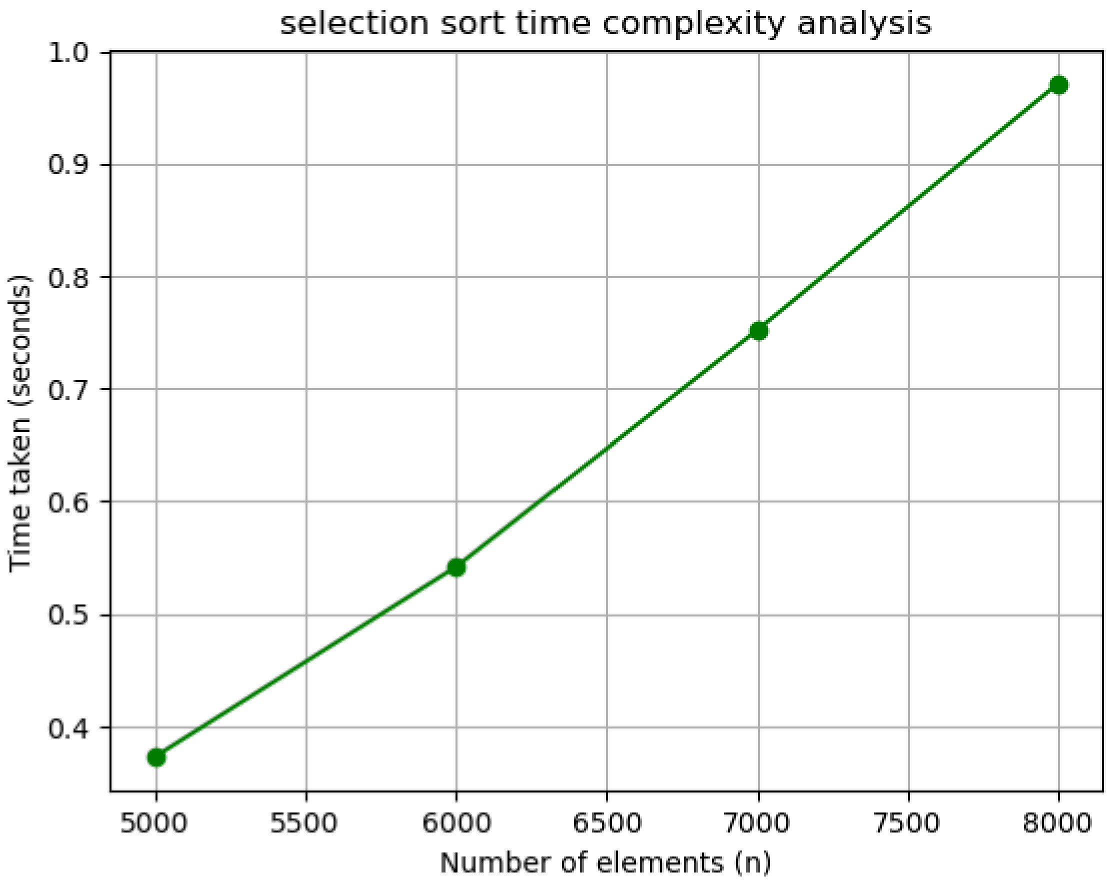
Title = Selection sort

In [1]:

```
import random
import time
import matplotlib.pyplot as plt

def selection_sort(arr):
    n= len(arr)
    for i in range(n):
        min_idx = i
        for j in range(i+1, n):
            if arr[j]< arr[min_idx]:
                min_idx = j
        arr[i], arr[min_idx] = arr[min_idx], arr[i]

    return arr

arr = list(map(int, input("Enter the list of elements separated by space: ").spl
print("Input array: ", arr)

sorted_arr = selection_sort(arr)
print("Sorted array:", sorted_arr)

start_time = time.time()
sorted_arr = selection_sort(arr)
end_time = time.time()
print("Time taken to sort:", end_time - start_time, "seconds")
print("========================================================")

n_values = [5000, 6000, 7000, 8000]
time_values = []

for n in n_values:
    arr = [random.randint(1, 9) for _ in range(n)]

    start_time = time.time()
    sorted_arr = selection_sort(arr)
    end_time = time.time()

    time_taken = end_time - start_time
    print("Time taken to sort", n, "elements:", time_taken, "seconds")
    time_values.append(time_taken)
print("===============================================================")

plt.plot(n_values, time_values, color = 'green', marker='o')
plt.xlabel('Number of elements (n)')
plt.ylabel('Time taken (seconds)')
plt.title('selection sort time complexity analysis')
```

```
plt.grid(True)
plt.show()
```

```
Input array:  [20, 30, 40, 50, 60, 40, 50]
Sorted array: [20, 30, 40, 40, 50, 50, 60]
Time taken to sort: 0.0 seconds
========================================================
Time taken to sort 5000 elements: 0.37334203720092773 seconds
Time taken to sort 6000 elements: 0.5418598651885986 seconds
Time taken to sort 7000 elements: 0.7518167495727539 seconds
Time taken to sort 8000 elements: 0.9708788394927979 seconds
===============================================================
```



In [ ]: